

Aminoglycosides

Thomas P. Lodise
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Short View Summary

- Usual intravenous (IV) adult dose for gram-negative infections: 5–7 mg/kg every 24 hours for gentamicin and tobramycin; 15 mg/kg for plazomicin; and 20–30 mg/kg every 24 hours for amikacin
- Usual dose for synergy against gram-positive infections: 3 mg/kg/day gentamicin every 8, 12, or 24 hours
- Dose reductions required in patients with chronic kidney disease and acute kidney injury (AKI) or dysfunction
- The plasma concentration–time curve to minimum inhibitory concentration (AUC/MIC) ratio is the pharmacokinetic/pharmacodynamic parameter most closely linked to bacterial killing and efficacy
- Therapeutic drug monitoring: not always necessary for brief administration (≤ 3 days); trough concentrations should be unmeasurable; typically, two serum concentrations in postdistribution phase (≥ 1 hour after administration, separated by ≥ 1.5 half-lives)
- Cerebrospinal fluid penetration: low
- Common adverse effects: AKI, ototoxicity
- Genotyping for the mitochondrial genome variant in *MT-RNR1*, m.1555A>G can help avoid use of aminoglycosides and prevent ototoxicity
- Drug-drug interactions: increased acute kidney risk with concomitant amphotericin B, cisplatin, cyclosporine, foscarnet, furosemide, IV radiocontrast agents, methoxyflurane, polymyxins, and vancomycin
- Indications: sepsis/septic shock; hospital-acquired/ventilator-associated pneumonia; complicated intraabdominal infections; urinary tract infections; neutropenic fever; acute exacerbation of pulmonary disease (cystic fibrosis patients only); bloodstream infections/IV catheter-related infections; infective endocarditis; multidrug-resistant tuberculosis; select nontuberculous mycobacterial infections; zoonotic infections (i.e., brucellosis, chronic forms of bartonellosis, tularemia, and yersiniosis); febrile neutropenia; surgical prophylaxis for patients undergoing genitourinary, gastrointestinal, or colorectal procedures; orthopedic surgery (antibiotic-impregnated cement); ocular infections (topical preparations); sexually transmitted infections; and select protozoal infections (e.g., amebiasis and cutaneous leishmaniasis)

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Aminoglycoside antibiotics have been used since the 1940s, when systematic screening of soil actinomycetes for the elaboration of antimicrobial substances yielded streptomycin. Produced by a species of *Streptomyces* (*S. griseus*), it was the first antibiotic in the aminoglycoside family (Table 24.1)¹ to be derived, directly or indirectly, from *Streptomyces* spp. (aminoglycosides with names ending in -mycin) or from *Micromonospora* spp. (aminoglycosides with names ending in -micin). Neomycin, kanamycin, and gentamicin are fermentation products with two or three chemical constituents. Amikacin, netilmicin, and others are semisynthetic derivatives of the natural product. Plazomicin is a semisynthetic aminoglycoside derived from sisomicin.² All aminoglycosides share similar physical, chemical, and pharmacologic properties. Four compounds—amikacin, gentamicin, tobramycin, and plazomicin—are currently approved by the US Food and Drug Administration (FDA) for the treatment of serious infections due to gram-negative or gram-positive bacteria.³⁻⁶

The advantages of the aminoglycoside class include rapid bactericidal activity, presence of a postantibiotic effect (PAE), synergy when combined with certain antibiotics, and well-defined pharmacokinetic (PK) profiles.⁷⁻¹¹ As polycationic molecules, aminoglycosides bind to anionic compounds on the cell membrane, such as lipopolysaccharides (LPS), and displace outer membrane divalent cations, leading to increased permeability of the membrane and uptake of aminoglycosides (i.e., self-promoted uptake). Displacement of outer membrane divalent cations by aminoglycosides also compromises the integrity of the cell wall, and it is postulated this contributes to the observed bactericidal action of aminoglycosides.¹²⁻¹³ After entering the cell, aminoglycosides bind to the 30S component of the prokaryotic ribosome.⁹ Affinity for the ribosomal target contributes to the rapid bactericidal action of aminoglycosides.⁹⁻¹⁴ Aminoglycoside activity is pH-dependent and is inhibited in acidic environments.⁷ In vitro studies have shown that aminoglycosides have a prolonged PAE, during which bacterial growth is suppressed for a further 3 to 8 hours after drug concentrations fall below the minimum inhibitory concentration (MIC).⁷⁻⁸⁻¹¹⁻¹⁵ Aminoglycosides are highly synergic with cell wall-active drugs (i.e., β -lactams or glycopeptides) in vitro. In preclinical pharmacokinetic/pharmacodynamic (PK/PD) infection models, concomitant use of an aminoglycoside with a β -lactam often enhances bacterial killing and prevents regrowth and resistance emergence to both the aminoglycoside and companion β -lactam.¹⁶⁻⁴⁵ The ratio of the maximum plasma drug concentration to the MIC ($C_{\max}/\text{MIC}$) ratio has long been considered the PK/PD driver for aminoglycosides, but more recent evidence suggest the area under the plasma concentration–time curve to MIC (AUC/MIC) ratio is the PK/PD parameter most closely linked to bacterial killing and efficacy.¹⁰⁻²⁹⁻³⁶⁻⁴⁶⁻⁴⁹

Aminoglycosides are active in vitro against both gram-positive and gram-negative pathogens, but they are primarily recommended for use in combination therapy for most infections.⁵⁰⁻⁶³ Three legacy aminoglycosides—amikacin, gentamicin, and tobramycin—are broadly active against *Staphylococcus* spp. (primarily gentamicin) and gram-negative bacteria including *Pseudomonas* spp. (primarily amikacin and tobramycin), *Citrobacter* spp., *Enterobacter* spp., *Escherichia coli*, *Klebsiella* spp., *Proteus* spp., and *Serratia* spp.³⁻⁵ Plazomicin, a newly approved aminoglycoside, is highly active in vitro against *Staphylococcus aureus*, including methicillin-resistant (MRSA) strains, and most Enterobacterales, including carbapenem-resistant Enterobacterales (CRE) and strains resistant to ≥ 1 aminoglycosides.⁶⁻⁶⁴⁻⁶⁵ Some aminoglycosides have useful activity against mycobacteria, paromomycin has activity against selected colonic protozoan pathogens, and spectinomycin, a related antibiotic, has activity against *Neisseria gonorrhoeae*.⁶²⁻⁶³⁻⁶⁶⁻⁶⁸

Although aminoglycosides are effective at bacterial cell killing and have a low potential for allergic reactions, their mechanism of action can also increase the potential for certain toxicities.⁶⁹⁻⁷² The affinity of aminoglycosides for receptors on the proximal renal tubular epithelial cells can lead to acute kidney injury (AKI).¹¹ In addition, ototoxicity (cochlear and vestibular) and, rarely, neuromuscular blockade are potential aminoglycoside-related side effects.⁹⁻¹⁴⁻⁷³ Since the mid-1990s, clinicians have generally adopted weight-based, extended-interval short-course dosing strategies for

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TABLE 24.1

Family of Aminoglycosides in Clinical Use

carbapenem-resistant Enterobacterales (CRE) and strains resistant to ≥ 1 aminoglycosides. ^{6, 64, 65} Some aminoglycosides have useful activity against mycobacteria, paromomycin has activity against selected colonic protozoan pathogens, and spectinomycin, a related antibiotic, has activity against *Neisseria gonorrhoeae*. ^{62, 63, 66-68}

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TABLE 24.1

Family of Aminoglycosides in Clinical Use

GENERIC NAME	PROPRIETARY NAME	SOURCE	YEAR REPORTED	CHEMISTRY
Streptomycin	None	<i>Streptomyces griseus</i>	1944	Unique central aminocyclitol ring
Neomycin	Mycifradin, Neobiotic	<i>Streptomyces fradiae</i>	1949	Roughly equal proportions of neomycin B and C
Kanamycin	Kantrex	<i>Streptomyces kanamyceticus</i>	1957	Mixture of 95% kanamycin A and 5% kanamycin B
Paromomycin	Humatin	<i>S. fradiae</i>	1959	Part of neomycin family
Spectinomycin	Trobicin	<i>Streptomyces spectabilis</i>	1961	Chemically distinct but closely related to aminoglycosides
Gentamicin	Garamycin	<i>Micromonospora purpurea</i> and <i>Micromonospora echinospora</i>	1963	Roughly equal proportions of gentamicin C ₁ , C _{1a} , and enantiomers C ₂ and C _{2a}
Tobramycin	Nebcin	<i>Streptomyces tenebrarius</i>	1967	Natural 3'-deoxy derivative of kanamycin B
Sisomicin ^a	Siseptin	<i>Micromonospora inyoensis</i>	1970	Dehydro analogue of gentamicin C _{1a}
Dibekacin ^{a, b}		<i>S. kanamyceticus</i>	1971	Dideoxy derivative of kanamycin B
Amikacin ^b	Amikin	<i>S. kanamyceticus</i>	1972	Semisynthetic derivative of kanamycin A

Kanamycin	Kanulex	<i>Streptomyces kanamyceticus</i>	1957	Mixture of 50% kanamycin A and 5% kanamycin B
Paromomycin	Humatin	<i>S. fradiae</i>	1959	Part of neomycin family
Spectinomycin	Trobicin	<i>Streptomyces spectabilis</i>	1961	Chemically distinct but closely related to aminoglycosides
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Amikacin ^b	Amikin	<i>S. kanamyceticus</i>	1972	Semisynthetic derivative of kanamycin A
Netilmicin ^b	Netromycin	<i>M. inyoensis</i>	1975	N-Ethyl derivative of sisomicin
Isepamicin ^{a, b}		<i>M. purpurea</i>	1978	I-N-S- α -Hydroxy B amino propionyl derivative of gentamicin B
Plazomicin	Zemdri	<i>M. inyoensis</i>	2018	Semisynthetic derivative of sisomicin

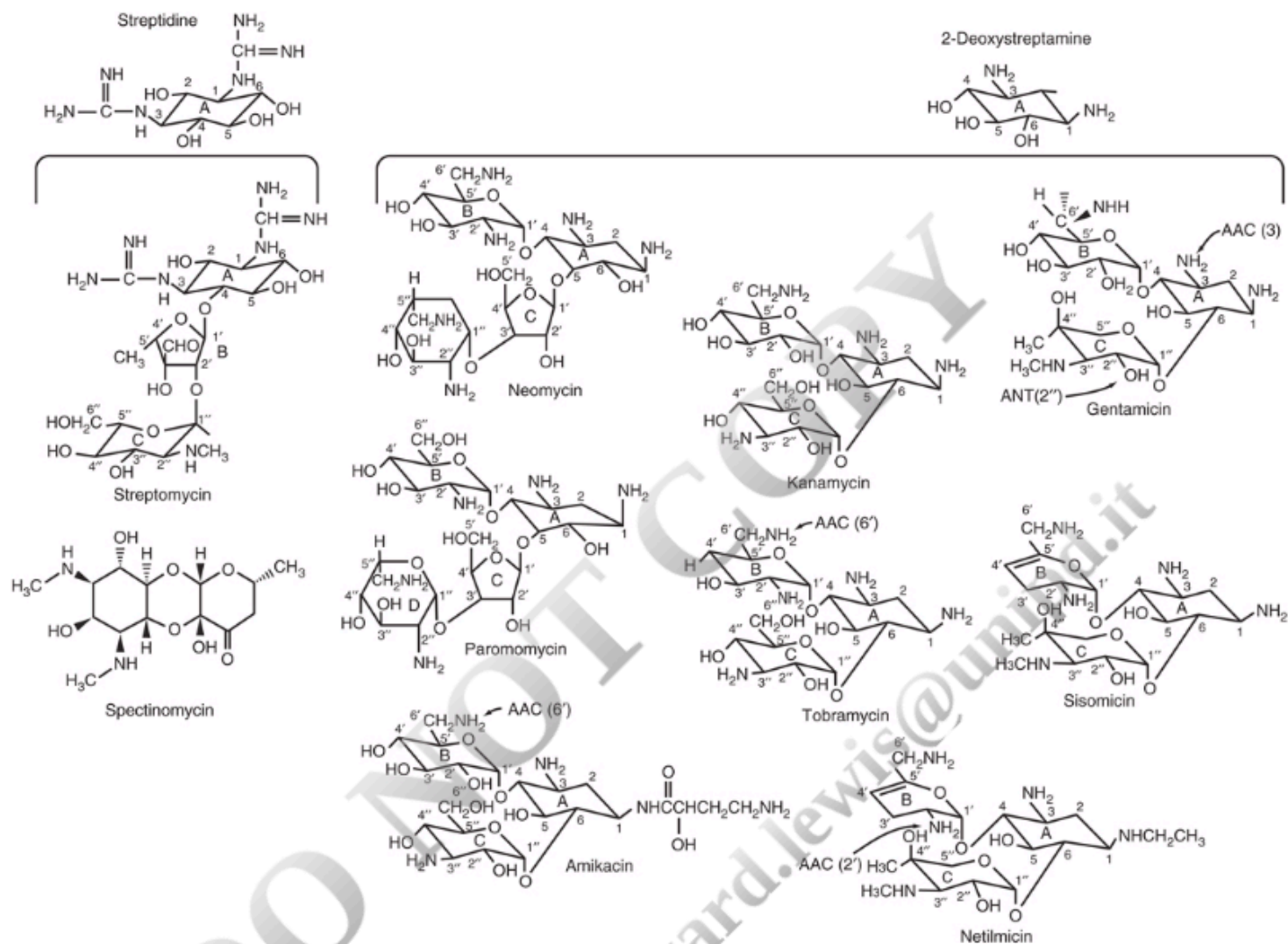
^a Approved for human use in countries other than the United States.

^b Semisynthetic aminoglycosides.

Modified from Wright GD, Berghuis AM, Mobashery S. Aminoglycoside antibiotics: structures, functions, and resistance. In: Rosen BP, Mobashery S, eds. *Resolving the Antibiotic Paradox: Progress in Understanding Drug Resistance and Development of New Antibiotics*. New York: Plenum; 1998.

Amid concerns about toxicity (particularly irreversible vestibular injury), the use of aminoglycosides began to decline in the 1980s in favor of newer antibiotic classes such as fluoroquinolones, which were perceived to be less toxic. ⁹ The 2021 National Safety Network Antimicrobial Use Option Report indicates that amikacin, tobramycin, and gentamicin account for 1% to 2% and 2% to 10% of adult and pediatric antibiotic prescriptions in the inpatient setting, respectively. ⁷⁶ However, the increasing prevalence of multidrug-resistant (MDR) gram-negative pathogens, including CRE, *Pseudomonas aeruginosa*, and *Acinetobacter* spp., and nontuberculous mycobacteria—for which therapeutic options are limited—has led to renewed interest in aminoglycosides for use as monotherapy or in combination with other antibacterials. ^{7, 66, 67, 72, 77} Although most studies have failed to demonstrate improved outcomes in patients treated with antibiotic combinations with an aminoglycoside relative to monotherapy, ^{78, 79} patients with septic shock receiving combinations including

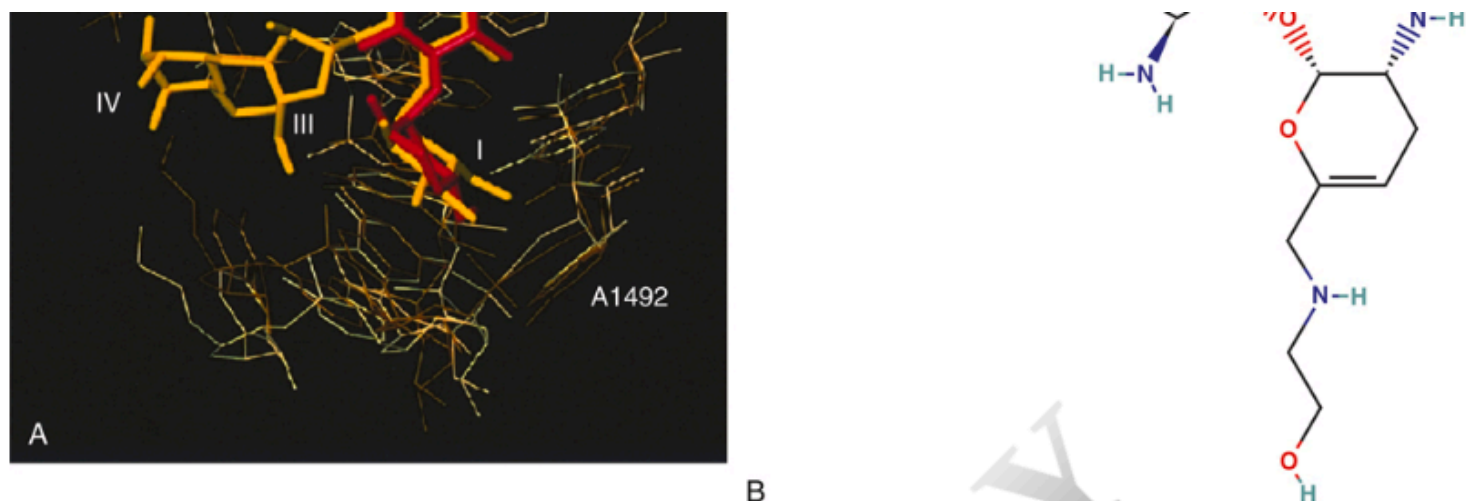
Description of FIG. 24.1



The annotated chemical diagram displays the chemical structures of various aminoglycoside antibiotics and spectinomycin, grouped by structural subtypes based on their aminocyclitol core rings. On the left, streptidine and its derivative streptomycin are shown, with ring labels and carbon positions marked. Streptomycin contains a streptidine ring linked to a second sugar ring and a third methylated sugar ring forming a trisaccharide structure. Spectinomycin, structurally distinct, is shown by a non-glycosidic aminocyclitol antibiotic with a unique tricyclic structure containing hydroxyl, methyl, and amino groups. The right and center sections of the image display aminoglycosides with a 2-deoxystreptamine core, including neomycin, kanamycin, gentamicin, paromomycin, tobramycin, sisomicin, and amikacin. Each molecule contains ring systems labeled A, B, and C, often with additional substituents or sugars. The image also identifies modification sites for resistance-conferring bacterial enzymes, including acetyltransferases AAC 3', AAC 2' and AAC 6' and the adenylyltransferase ANT 2'. Specific structures like gentamicin and amikacin are shown with varied R-group substitutions representing different isomers, like gentamicin C 1, C 1 a, and C 2.

[Navigate back to FIG. 24.1](#)

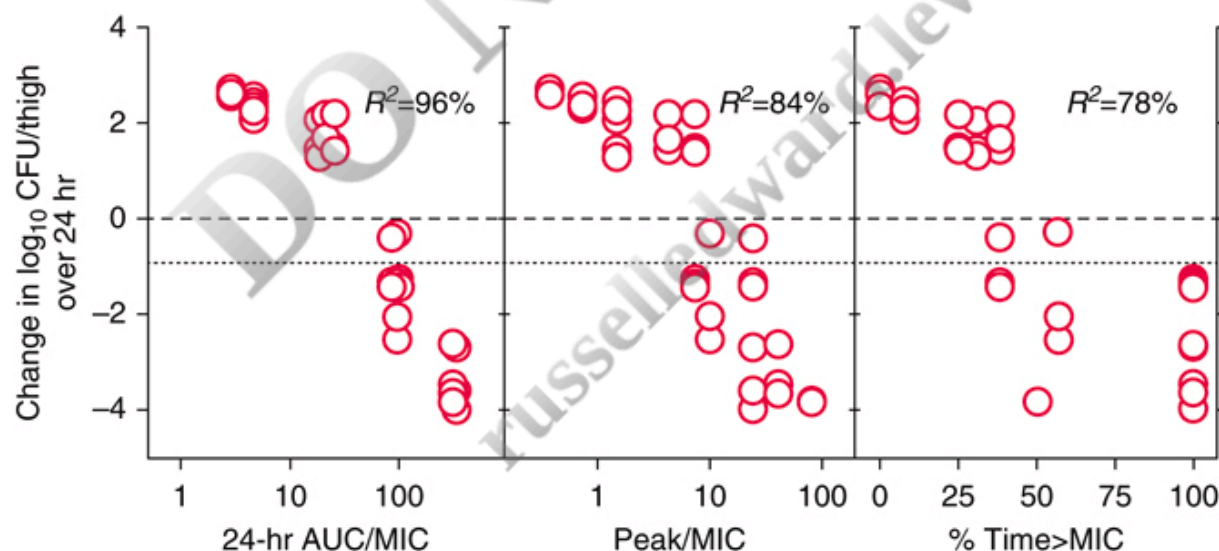
Description of FIG. 24.2



The illustration contains two annotated panels illustrating the structural interactions and chemical composition of aminoglycoside antibiotics. Panel A presents a three-dimensional molecular rendering of paromomycin and gentamicin C1a bound to segments of Escherichia coli 16S ribosomal RNA. The RNA is shown in wireframe form, representing different structural regions. Key ribosomal nucleotides including G1405, U1495, and A1492 are labeled. Both antibiotics share rings 1 and 2, which align identically in their respective binding pockets. In contrast, additional ring systems—rings 3 and 4 in paromomycin and ring 3 in gentamicin—show distinct orientations and interactions with surrounding RNA. Panel B displays the two-dimensional chemical structure of plazomicin. The structure consists of multiple amino sugar rings with defined stereochemistry and substitutions.

[Navigate back to FIG. 24.2](#)

Description of FIG. 24.3



The annotated scatter plot displays the correlation between three pharmacokinetic bar pharmacodynamic indices and the change in bacterial load, measured log base 10 colony-forming units per thigh, over 24 hours in neutropenic mice. The graph is divided into three adjacent sections, each representing a different PK bar PD index: 24-hour area under the concentration-time curve to minimum inhibitory concentration ratio, peak drug concentration to MIC ratio, and percentage of time drug concentration remains above MIC. Each section plots open circles indicating experimental data points for individual observations. The vertical axis, common to all three panels, ranges from plus 4 to minus 4 in log base 10 CFU

Page 326 analyzed as part of the Cochrane Library Review. ⁵⁰¹ There were no significant differences between inhaled aminoglycosides and placebo with regards to treatment-related adverse events of interest (i.e., auditory impairment, pneumothorax, and hemoptysis). However, tinnitus and voice alteration were significantly more common in individuals who received inhaled aminoglycosides.

No clinically meaningful differences in outcomes, resistance emergence, and safety were observed in studies that compared the long-term use of different inhalational formulations of aminoglycosides (i.e., tobramycin inhalation powder, tobramycin for inhalation solution [TOBI], inhaled tobramycin [intravenous preparation], and amikacin liposome inhalation suspension) in individuals with cystic fibrosis. ⁵⁰¹ Similarly, few differences were observed in long-term comparative studies between aerosolized aminoglycosides and other classes of antibiotics. ⁵⁰¹ In one open-labeled study ($n = 268$) of cystic fibrosis individuals who received aztreonam lysine for inhalation or tobramycin for inhalation solution, individuals who received aztreonam lysine for inhalation had a higher mean improvement in lung function and required less receipt of additional antibiotics. ⁵⁰² In another study that compared levofloxacin for inhalation solution versus tobramycin for inhalation solution, 17.5% of participants in the levofloxacin group were hospitalized compared to 28% of participants in the tobramycin group (RR, 0.62; 95% CI, 0.40 to 0.98). ⁵⁰³ Treatment-related adverse events were relatively infrequent across all antibiotic classes but were reported less in individuals who received inhaled tobramycin relative to those who received aztreonam lysine or colistimethate. ⁵⁰¹ ⁵⁰²

Aerosolized aminoglycosides have shown promise in chronic bronchiectatic infections but none are presently approved by the FDA for this indication. ⁵⁰⁴⁻⁵⁰⁷ Aerosolized aminoglycosides also are regularly used in patients with VAP but none is approved for use for this indication. Despite the lack of FDA approval, the HAP/VAP expert guidelines recommend use of aerosolized aminoglycosides or colistin, in combination with systemic agents, for patients with HAP/VAP caused by gram-negative organisms only susceptible to aminoglycosides or polymyxins. ⁵⁰ A meta-analysis presented in the IDSA/ATS guidelines identified a significant improvement in clinical cure rate with the addition of inhaled antibiotics for the treatment of MDR pathogens, though no differences in mortality were identified. In contrast, the IDSA Guidance on the Treatment of Antimicrobial Resistant Gram-Negative Infections panel does not suggest the use of nebulized antibiotics for the treatment of respiratory infections caused by resistant gram-negative pathogens due to the lack of benefit observed in clinical trials, concerns regarding unequal distribution in infected lungs, ⁵⁰⁸ and concerns for respiratory complications such as bronchoconstriction in patients receiving aerosolized antibiotics. ⁵⁷ ⁵⁸ Lastly, there are mounting data with inhaled amikacin (parenteral formulation or liposome inhalation suspension) for multidrug regimen for NTM pulmonary disease. However, inhaled amikacin is currently not recommended for inclusion in the initial regimen at this time, and its use should be reserved to patients with MAC pulmonary disease who have failed therapy after at least 6 months of guideline-based therapy. ⁶²

Prophylaxis

Clinical practice guidelines for antimicrobial prophylaxis in surgery suggest gentamicin or tobramycin, 5 mg/kg IV, as an alternative in patients with β -lactam allergy undergoing genitourinary and gastrointestinal procedures. ⁵⁰⁹ For patients undergoing cardiac or vascular surgeries or those with valvular heart disease, prophylaxis with an aminoglycoside is no longer recommended. For patients undergoing colorectal procedures, a mechanical bowel preparation combined with oral neomycin sulfate plus oral erythromycin base or with oral neomycin sulfate plus oral metronidazole is often used with IV prophylaxis. ⁵⁰⁹ The risk of infection after elective colorectal procedures was significantly reduced by mechanical cleansing of the bowel plus oral administration of neomycin and erythromycin or metronidazole in addition to standard intravenous antibiotic prophylaxis in controlled trials. ⁵¹⁰ ⁵¹¹ A controlled trial demonstrated that an oral selective digestive decontamination containing gentamicin effectively eradicated carbapenem-resistant *K. pneumoniae* gastrointestinal carriage ⁵¹² and may be used in nosocomial outbreaks. ⁵¹³ The safety and efficacy of topical gentamicin in cardiac surgery

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randomized trial ⁵¹⁵ have demonstrated reduced catheter-related infections, but emergence of resistant pathogens remains a concern. ⁵¹⁶

Aminoglycosides in Orthopedic Surgery

Antibiotic-impregnated cement is used with increasing frequency in primary hip and knee arthroplasties and revision procedures of infected total joint arthroplasties, for which the FDA has approved premixed aminoglycoside in bone cement products. ⁵¹⁷ Incorporation of larger amounts of antibiotic, usually gentamicin or tobramycin, allows release of higher drug concentrations but may adversely affect mechanical properties. Concentrations and properties vary among producers. ⁵¹⁸⁻⁵²³ Persistence of bacterial growth as adherent biofilms remains a potential problem, ¹⁶⁸ ⁵¹⁸ ⁵²⁴ and AKI has been reported. ⁵²⁵ ⁵²⁶ To date, one multicenter evaluation failed to show reduction in the risk of infection. ⁵²⁷ Additionally, a review of 20 mostly uncontrolled reports of antibiotic-containing spacers did not allow evaluation of whether such adjunctive therapy provided additional benefits to systemic antibiotic therapy. ⁵²⁸ Prospective trials in aminoglycoside-containing beads for established osteomyelitis and prosthetic joint-associated infections are needed. ⁵²⁹

Other Clinical Uses

Aminoglycosides are commercially available as topical ophthalmic preparations (e.g., ointments or suspensions) and are frequently used, alone and in combination, to treat patients with ocular infections. ⁵³⁰ Oral paromomycin has been employed as an alternative therapy for infection caused by *Entamoeba histolytica* and ulcerative *Leishmania major*. ⁵³¹⁻⁵³⁴ Intravesicular gentamicin has been investigated, with anecdotal success, for recurrent UTIs in intermittently catheterized patients. ⁵³⁵ Spectinomycin was an alternative agent used intramuscularly in the treatment of *N. gonorrhoeae*, ⁵³⁶ but it is no longer distributed in the United States. It was previously recommended for pregnant women, patients in areas with a high prevalence of fluoroquinolone resistance, and men who have sex with men. It showed 98% efficacy in uncomplicated urogenital and anogenital gonococcal infections but only 52% efficacy in gonococcal infections involving the pharynx, where it does not reach therapeutic concentrations. ⁵³⁷ Spectinomycin is also not effective in the treatment of infections with *Treponema pallidum* or *Chlamydia trachomatis*. Currently, spectinomycin is an alternative therapy for patients who are allergic to β -lactams and for patients who are infected with resistant strains of gonococci. Alternatively, gentamicin has been used to treat urogenital gonococcal infection. ⁵³⁸

Individualized Aminoglycoside Dosing

Aminoglycosides have traditionally been dosed on weight (mg/kg) and an estimate of the kidney function (eCrCL or eGFR). ¹⁰ Suggested initial dosing regimens for typical aminoglycosides in clinical use are listed in [Table 24.5](#). Higher initial doses of amikacin (20–30 mg/kg) have been suggested with further dose adjustment after therapeutic drug monitoring (TDM), but consensus is needed. ⁵³⁹ Given the age and generic status of this drug class, high-quality clinical studies are unfortunately unlikely to be performed to validate these doses apart from plazomicin, which is the most contemporary addition to the armamentarium. ³⁴⁵ However, the lack of alternate therapeutic agents for MDR gram-negative pathogens has led individual institutions to adopt these higher doses. ⁵⁴⁰ ⁵⁴¹

One out of two adults in the United States will be obese by 2030, which impacts aminoglycoside dosing considerations. ⁵⁴² Use of actual weight in obese patients can lead to overexposure. ⁵⁴³ To prevent this, the most common approach is to use adjusted body weight, which is ideal body weight plus 40% of the difference between actual and ideal body weight in obese patients. ⁵⁴³ The typical definition of obesity for this use of adjusted body weight is when actual weight exceeds ideal body weight by 25% to 30% or if the body mass index ≥ 30 kg/m². ⁵⁴⁴ More recent work indicates that lean body weight permits simplified aminoglycoside dosing across all weight strata, with clearance best predicted by the

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In resource-poor settings where TDM may be unavailable or delayed, an estimate of kidney function will be more informative than weight to define the daily dose. ⁵⁴⁴ As noted, aminoglycoside clearance is a close surrogate of the measured glomerular filtration rate. Creatinine clearance overestimates GFR by 10% to 15% as it includes tubular secretion (an elimination process not applicable to aminoglycosides). In contrast, use of the most contemporary GFR equations based on serum creatinine (or cystatin C when available) can be translated to an estimate of aminoglycoside clearance (CL). ⁵⁴⁵ An AUC target exposure multiplied by this CL estimate approximates the daily dose that is necessary in the individual patient. ⁵⁴⁴ Although this bedside approach is a “best guess,” in the absence of resources to measure aminoglycoside concentrations, use of estimated kidney function is much better than body weight to estimate the daily dose.

In patients with kidney failure who require kidney replacement therapies such as dialysis, the dosing of aminoglycosides will depend on the characteristics of the dialysis membrane, duration of dialysis, the patient’s blood pressure during dialysis, and other variables. ^{548–550} Patients in sepsis who may be receiving continuous hemofiltration or dialysis the clearance of aminoglycosides is 80% to 90% of the effluent rate or equivalent to a CrCl of 10 to 50 mL/minute. ^{550–552} For patients on hemodialysis, a traditional dose is given every 48 to 72 hours, and on the day of hemodialysis an additional one-half of the full dose is given (if the traditional dose was administered before dialysis) after dialysis to replace drug that was removed by the dialysis or simply dosed after dialysis. Because of individual variability, serum levels should be measured. ^{553, 554}

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Therapeutic Drug Monitoring Approach

If therapy is expected to continue for ≥ 3 days and TDM is available, multiple possible approaches exist with varying levels of precision and complexity as ranked below (highest to lowest precision) that are resource dependent:

- Measurement of two concentrations and use of Bayesian software (software using two concentrations to gauge)
- Measurement of two concentrations and use of PK equations (linear, so calculate)
- Measurement of one concentration and use of Bayesian software
- Measurement of one concentration (random) and use of a dosing nomogram
- Measurement of two concentrations and interpretation using concentration thresholds (trough and peak)
- Measurement of one concentration (peak or trough) and interpretation using concentration thresholds

It is unclear whether every patient administered an aminoglycoside requires an individual PK evaluation, ⁵⁵⁷ but

